

Experiment 14: House Price Prediction using Machine Learning

Aim

To implement a **House Price Prediction Model** using the **Linear Regression** algorithm in Python.

Python Program

```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load California Housing Dataset
housing = fetch_california_housing()

X = housing.data
y = housing.target

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Linear Regression Model
model = LinearRegression()

# Train Model
model.fit(X_train, y_train)

# Predict House Prices
y_pred = model.predict(X_test)

print("First 10 Actual Prices:")
print(y_test[:10])
```

```

print("\nFirst 10 Predicted Prices:")
print(y_pred[:10])

# Performance Evaluation
print("\nMean Squared Error:")
print(mean_squared_error(y_test, y_pred))

print("\nR2 Score:")
print(r2_score(y_test, y_pred))

# Predict one house
sample_house = [X_test[0]]

predicted_price = model.predict(sample_house)

print("\nPredicted House Price:")
print(predicted_price[0])

```

Sample Output

```

... First 10 Actual Prices:
[0.477  0.458  5.00001 2.186  2.78  1.587  1.982  1.575  3.4
 4.466 ]

First 10 Predicted Prices:
[0.72604907 1.76743383 2.71092161 2.83514727 2.60695807 2.01073856
 2.64067386 2.16706161 2.74012056 3.90361526]

Mean Squared Error:
0.5305677824766757

R2 Score:
0.595770232606166

Predicted House Price:
0.7260490726243773

```